

ReLU#

<https://pytorch.org/docs/stable/generated/torch.nn.ReLU.html#torch.nn.ReLU>

Description:

Applies the rectified linear unit function element-wise. $\text{ReLU}(x) = \max(0, x)$

$\text{ReLU}(x) = (x)^+ = \max(0, x)$

inplace (bool) – can optionally do the operation in-place. Default: False

Input: $(*)^{(*)} (*)$, where $*^*$ means any number of dimensions. Output: $(*)^{(*)} (*)$, same shape as the input.

Function Signature

```
class torch.nn.ReLU(inplace=False)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.ReLU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

An implementation of CReLU - <https://arxiv.org/abs/1603.05201>

```
>>> m = nn.ReLU()  
>>> input = torch.randn(2).unsqueeze(0)  
>>> output = torch.cat((m(input), m(-input)))
```

Detailed Documentation

Documentation

Applies the rectified linear unit function element-wise.

$\text{ReLU}(x) = \max(0, x)$

`inplace (bool)` – can optionally do the operation in-place. Default: `False`

Input: $(*)^{(*)} (*)$, where $***$ means any number of dimensions.

Output: $(*)^{(*)}(*)$, same shape as the input.

Return the extra representation of the module.

Runs the forward pass.

Applies the rectified linear unit function element-wise.

$\text{ReLU}(x) = \max(0, x)$

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